

Tracy (Huiwen) Han

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EDUCATION

Boston University

B.A. in Biology; minors in Computer Science and Business Administration; GPA 3.7/4.0

Boston, MA

2020 – 2023

PUBLICATIONS

- [1] **Huiwen Han***, Ziyao Zeng, Zihao Zhu, Sisong Bei, Hao Wang, Hongyu Xu, Yan Han, Degui Zhi, and Zhiwen Fan. “Contextual Credit Policy Optimization,” under review, 2026.
- [2] **Huiwen Han***, Lulin Liu*, Bangya Liu, Yuanhao Cai, Nuo Chen, Xiaoqing Wang, Ziqian Xie, Chenyu You, Shuiwang Ji, Degui Zhi, and Zhiwen Fan. “AnaDiffusion: Anatomically Compositional Latent Diffusion for Controllable 3D Brain MRI Generation.” In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. arXiv:2608.23014.
- [3] **Huiwen Han***, Adam R. Pines, Yixin Wang, Yu Zhang, and Christopher W. Austelle. “NetHGL: Hierarchical Graph Learning Identifies Remission-Associated Network Organization in Treatment-Resistant Depression Following Stanford Neuromodulation Therapy.”
- [4] Minseong Kweon, Wenyuan Zhao, Nuo Chen, Lulin Liu, **Huiwen Han**, Zihao Zhu, Srinivas Shakkottai, Chao Tian, and Zhiwen Fan. “DarkVGGT: Seeing Through Darkness Using Thermal Geometry without Daylight Tax.” In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. arXiv:2606.11326.
- [5] Tian Xia*, Nuo Chen*, Zihao Zhu, **Huiwen Han**, Ziqian Xie, Zhiwen Fan, and Degui Zhi. “Surface-volume self-supervised representation learning of brain MRI for genetic discovery.” arXiv:2610.02114, 2026.
- [6] Kenneth Shinozuka, Clayton Olash, **Huiwen Han**, Azeezat Azeez, Malvika Sridhar, Andrew D. Geoly, Catherine Daye, Saron Hunegnaw, Kirsten N. Cherian, Jakob N. Keynan, Randi E. Brown, Derrick M. Buchanan, John P. Coetzee, Ian H. Kratter, Raag D. Airan, Martijn Arns, Maheen M. Adamson, Manish Saggarr, and Camarin Rolle. “Dynamic Substates of the Default Mode Network Are Associated with Mystical Experiences and Clinical Outcomes after Magnesium-Ibogaine Treatment in Veterans with Traumatic Brain Injury.” medRxiv, 2026. doi:10.64898/2026.07.20.26358512.
- [7] Florence X. Doo, **Huiwen Han**, Bradley A. Maron, and Heng Huang. “Trustworthy Clinical Thinking in MLLMs: Hierarchical Energy-Based Reasoning for Interpretable Medical Scans (HERMES).” In *AI for Clinical Applications*, LNCS, Springer, 2025. Accepted for an **oral presentation** at the 1st Workshop on Multimodal Large Language Models in Clinical Practice, 28th International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2025.

RESEARCH EXPERIENCE

Brain Stimulation Lab, Stanford University

Research Assistant; Advisors: Christopher Austelle and David Spiegel

Stanford, CA

Sep. 2024 – Present

- Developed [multimodal clinical-trial data infrastructure](#) that automates reproducible, auditable workflows for high-risk clinical data; the resulting environment supports tiered access control and lays the groundwork for an agentic research system. Funded by a \$20,000 Stanford Rigor & Reproducibility grant.
- Model neural and pathophysiological mechanisms of psychiatric neuromodulation from neuroimaging and clinical assessment data using probabilistic and deep learning methods (e.g., HMMs, GNNs, and HSSMs); identified a candidate treatment-response biomarker with hierarchical graph neural networks.
- Administer brain stimulation treatments; collect and analyze multimodal clinical data, including neuroimaging, biometrics, and actigraphy, for psychiatric neuromodulation trials.

Physical Automation Intelligence Lab, Texas A&M University

Visiting Student; Advisor: Zhiwen Fan

College Station, TX

Jun. 2025 – Present

- Developed Contextual Credit Policy Optimization (CCPO), a critic-free reinforcement learning method for long-horizon LLM agents that assigns dual contextual step-level credit from historical returns and future progress; future credit stabilizes learning as the historical signal diminishes over training, and CCPO substantially reduces repetitive agent behaviors for efficient multi-step task execution (under review, 2026).
- Developed compositional latent-diffusion models for controllable 3D brain MRI generation; AnaDiffusion improves anatomical realism when generating 3D brains across Alzheimer’s disease, mild cognitive impairment, and healthy aging (NeurIPS 2026).
- Designed evidence-grounded agents, e.g., a lunar olivine science agent with the Planetary Science Institute and a multimodal agent that uses video for longitudinal behavioral profiling and symptom monitoring in children with autism spectrum disorder (ASD).

Wheelock Lab, Washington University in St. Louis

Student Research Assistant; Advisor: Muriah D. Wheelock

St. Louis, MO

Jun. 2024 – Aug. 2025

- Developed interpretable GNNs for decoding neural dynamics during sleep across human developmental stages.

Harvard T.H. Chan School of Public Health

Student Research Assistant; Advisors: William Hsiao and Winnie Yip

Boston, MA

Nov. 2021 – May 2023

- Analyzed health-system mechanisms related to population aging, electronic health records, and procurement.

INDUSTRY EXPERIENCE

Independent Contractor

2026 – Present

- Engineer AI training data across domains, including data for training coding agents and domain-specialized synthetic data; built agentic loops and self-evolving skills that automate data generation.

Parse Biosciences

Multi-omics Specialist

Seattle, WA

Jun. 2023 – May 2024

- Advised research labs and organizations on multi-omic single-cell sequencing design and workflows; presented on scalable single-cell sequencing at NIH and USDA campuses and at Neuroscience 2023, the Society for Neuroscience (SfN) annual meeting.

RESEARCH SERVICE

Reviewer for *Transactions on Machine Learning Research*.Reviewer for *IEEE Transactions on Circuits and Systems for Video Technology*.Co-reviewer for *Psychiatry and Clinical Neurosciences*.Co-reviewer for *CNS Neuroscience & Therapeutics*.